In VADER (Valence Aware Dictionary and sEntiment Reasoner), the **compound score** is the overall sentiment score for a piece of text. It ranges from **-1** (most negative) to **+1** (most positive) and is a normalized score calculated based on the sentiment intensity of words in the text.

Here's a step-by-step breakdown of how the **compound score** is calculated:

1. **Sentiment Lexicon**

VADER uses a sentiment lexicon where each word in the text is assigned a pre-determined sentiment score between **-4** (most negative) and **+4** (most positive). Words like "excellent" or "love" have high positive scores, while words like "terrible" or "hate" have high negative scores.

2. **Summing Sentiment Scores**

The individual sentiment scores of each word in the text are summed up to create a **raw score** for the text. For example, if a sentence contains words with the following sentiment scores:

- "love" = +3.0
- "this" = 0.0
- "movie" = 0.0
- "terrible" = -2.5

The sum of sentiment scores would be:
$+3.0 + 0.0 + 0.0 - 2.5 = +0.5$

3. **Adjustments for Punctuation and Capitalization**

VADER adjusts the sentiment scores to account for context and emphasis:

- **Exclamation Marks**: Each exclamation mark adds a small boost to the sentiment intensity.
- **Capitalization**: Words in all caps (e.g., "LOVE") are given more weight in the calculation, as they are considered emphasized.
- **Negations**: Words like "not" or "never" can flip the sentiment of a word from positive to negative or reduce its intensity.

4. **Normalization**

After summing the adjusted sentiment scores, VADER normalizes this value to bring it within the range of **-1 to +1**. The normalization is done using the formula:

$$\text{compound} = \frac{\text{sum of sentiment scores}}{\sqrt{(\text{sum of sentiment scores})^2 + \alpha}}$$

Where **α** is a constant, usually set to 15, that helps smooth the score. The normalization ensures that the compound score is bounded between **-1** and **+1**, allowing for easier interpretation of the overall sentiment.

- **Positive compound score**: Indicates overall positive sentiment.
- **Negative compound score**: Indicates overall negative sentiment.

- **Compound score close to 0**: Indicates neutral sentiment.

Example of Compound Score Calculation:

Let's break down an example:

Text: "I absolutely love this movie! It's fantastic but the ending was terrible."

Step 1: Assign sentiment scores

- "absolutely" = 1.5 (positive intensifier)
- "love" = 3.0 (positive)
- "fantastic" = 3.5 (positive)
- "terrible" = -2.5 (negative)

Step 2: Sum sentiment scores

$$1.5 + 3.0 + 3.5 - 2.5 = 5.5$$

Step 3: Adjust for exclamation marks and capitalization (if present)

Step 4: Normalize the score using the formula

$$\text{compound} = \frac{5.5}{\sqrt{5.5^2 + 15}} = \frac{5.5}{\sqrt{30.25 + 15}} = \frac{5.5}{\sqrt{45.25}} = \frac{5.5}{6.73} \approx 0.82$$

The **compound score** for this text would be approximately **+0.82**, indicating an overall positive sentiment.

Conclusion:

- The **compound score** in VADER is a normalized sum of individual word sentiment scores.
- It adjusts for context like negations, punctuation, and capitalization.
- The final score ranges from **-1** (most negative) to **+1** (most positive), with scores closer to 0 indicating neutral sentiment.